Finite-field basis obstructions for zero link Turán density

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Abstract

For an r-uniform hypergraph H, let H_k be obtained by adjoining the same k-vertex stem to every edge, and let $\pi_\infty(H) = \lim_{k\to\infty} \pi_{r+k}(H_k)$. We formulate a finite homomorphism obstruction that forces $\pi_\infty(H) > 0$. For a prime power q and $m \geq 0$, let $B_{q,m}(r)$ have vertex set $\mathbb{F}_q^{r+m}$ and edges the linearly independent r-sets. If H has no homomorphism to $B_{q,m}(r)$, then

$$\pi_\infty(H) \geq \prod_{j=m+1}^{\infty} (1 - q^{-j}) > 0.$$

We apply the criterion to a six-vertex three-graph and its dual. Every vertex link of either three-graph is tripartite, but both have link Turán density at least $\prod_{j\geq 1}(1-2^{-j}) \approx 0.288788$. Thus tripartiteness of every vertex link is not sufficient for zero link Turán density.

1 Suspensions and basis hypergraphs

Let H be a finite nonempty r-uniform hypergraph, $r \geq 2$. For $k \geq 0$, its k-fold suspension H_k is the $(r+k)$-uniform hypergraph obtained by adding the same new k vertices to every edge. Define

$$\pi_\infty(H) = \lim_{k\to\infty} \pi_{r+k}(H_k).$$

The limit exists because the sequence is nonincreasing: every vertex link in an H_{k+1}-free $(r+k+1)$-graph is H_k-free, and averaging links gives

$$\pi_{r+k+1}(H_{k+1}) \leq \pi_{r+k}(H_k).$$

For a prime power q and $m \geq 0$, let $B_{q,m}(r)$ be the r-graph on $\mathbb{F}_q^{r+m}$ whose edges are the linearly independent unordered r-sets. The zero vector is therefore an isolated vertex of $B_{q,m}(r)$; deleting it does not change any edge or any homomorphism from a nonempty edge of H.

Theorem 1. *If there is no hypergraph homomorphism*

$$H \longrightarrow B_{q,m}(r),$$

then

$$\pi_\infty(H) \geq \beta_{q,m} := \prod_{j=m+1}^{\infty} (1 - q^{-j}) > 0. \tag{1}$$

Consequently, $\pi_\infty(H) = 0$ *implies that* H *admits a homomorphism to* $B_{q,0}(r)$ *for every prime power* q.

2 Proof of the obstruction theorem

Set $R = r + k$. On $\mathbb{F}_q^{R+m} \setminus \{0\}$, let $G_{R,m}$ be the R-uniform hypergraph of linearly independent R-sets. It has

$$|E(G_{R,m})| = \frac{1}{R!} \prod_{i=0}^{R-1} (q^{R+m} - q^i)$$

edges. A balanced blow-up of $G_{R,m}$ has limiting edge density

$$\begin{aligned} d_{R,m}(q) &= \frac{R!|E(G_{R,m})|}{(q^{R+m} - 1)^R} \\ &= \frac{\prod_{j=m+1}^{m+R}(1 - q^{-j})}{(1 - q^{-(R+m)})^R}. \end{aligned} \tag{2}$$

The denominator in (2) tends to one, and the numerator tends to $\beta_{q,m}$. The infinite product is positive because $\sum_{j \geq m+1} q^{-j} < \infty$.

It remains to show that the blow-ups are H_k-free. Suppose a copy of H_k in a blow-up is compressed to the vertex classes of $G_{R,m}$. The common stem lies in every suspended edge; the image of any one suspended edge is a linearly independent R-set, so its k stem vectors are linearly independent. Let W be their span. Quotienting by W sends the image of every original H-edge to r independent vectors in

$$\mathbb{F}_q^{R+m}/W \cong \mathbb{F}_q^{r+m}.$$

This gives a homomorphism $H \to B_{q,m}(r)$, contrary to the hypothesis. Nonadjacent vertices may be identified, which is precisely why the statement uses homomorphisms; isolated vertices may be mapped arbitrarily.

Thus $\pi_{r+k}(H_k) \geq d_{r+k,m}(q)$ for every k. Taking the limit in (2) proves (1).

3 A six-vertex application

Let the vertex set be

$$\{c\} \sqcup A \sqcup B, \qquad |A| = 3, \quad |B| = 2.$$

Define the three-graph H_* by the eight edges

$$\{c, a, b\} \quad (a \in A,\ b \in B), \qquad \{c\} \cup B, \qquad A.$$

Proposition 1. *There is no homomorphism $H_* \to B_{2,0}(3)$.*

Proof. Suppose ϕ were a homomorphism and put $x = \phi(c)$. Modulo $\langle x \rangle$, the two vertices of B occupy two distinct nonzero points of the two-dimensional quotient, because $\{c\} \cup B$ maps to a basis. Every edge $\{c, a, b\}$ forces the image of each $a \in A$ to avoid those two quotient points. The projective line over $\mathbb{F}_2$ has only three points, so all three A-images lie over the remaining point and hence inside one two-dimensional subspace $\langle x, u \rangle$. They cannot form a basis, contradicting that A is an edge. □

Theorem 1 gives

$$\pi_\infty(H_*) \geq \prod_{j=1}^{\infty} (1 - 2^{-j}) \approx 0.288788. \tag{3}$$

Every vertex link of H_* is tripartite. The link at c is $K_{3,2}$ together with the edge on B, with parts $A, \{b_1\}, \{b_2\}$. A link at a vertex of A is the disjoint union of a two-leaf star and an edge, and a link at a vertex of B is a star.

The dual three-graph H_*^* has edges A, $\{c\} \cup B$, and $A' \cup \{b\}$ for every $b \in B$ and every two-set $A' \subset A$. If the images of A form the basis u_1, u_2, u_3, then the image of either $b \in B$ must have all three coordinates nonzero, hence equals $u_1 + u_2 + u_3$. The two B vertices then have the same image, contradicting that $\{c\} \cup B$ maps to a basis. Thus the same lower bound (3) holds for H_*^*. Its links are also tripartite by direct inspection.

Neither H_* nor H_*^* is the basis hypergraph of a matroid. In H_*, the edges $X = \{c, a_1, b_1\}$ and $Y = A$ violate basis exchange at $c \in X \setminus Y$; duality gives the corresponding statement for H_*^*.

4 Scope and provenance

The parameter π_∞ and the question of which three-graphs have zero link Turán density are stated in Calbet's problem in *Problems from BCC30*, Section 30.5 [1]. Finite-field independent-set hosts, their Euler-product densities, and their use for daisies occur in Ellis–Ivan–Leader, especially Theorem 4 [2]. The relation between suspension, contraction and matroid basis hypergraphs is developed by van der Pol–Walsh–Wigal in Proposition 2.2, Corollary 2.3 and Section 7.1 [3].

The contribution claimed here is limited to the homomorphism formulation for arbitrary finite uniform H and to the pair H_*, H_*^*, which pass every local tripartite-link test but still have positive link Turán density. The cited ingredients and their matroid and daisy applications are not claimed as new.

This does not classify the three-graphs of zero link Turán density, and the general formulation may still be recognizable to specialists as a folklore abstraction of the finite-field blow-up construction.

References

[1] P. J. Cameron (editor), *Problems from BCC30*, Section 30.5, problem of A. Calbet, arXiv:2409.07216. arXiv:2409.07216.

[2] D. Ellis, M.-R. Ivan and I. Leader, *Turán Densities for Daisies and Hypercubes*, Theorem 4, arXiv:2401.16289v6. arXiv:2401.16289.

[3] J. van der Pol, Z. Walsh and M. C. Wigal, *Turán densities for matroid basis hypergraphs*, Proposition 2.2, Corollary 2.3 and Section 7.1, arXiv:2502.03673v2. arXiv:2502.03673.

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